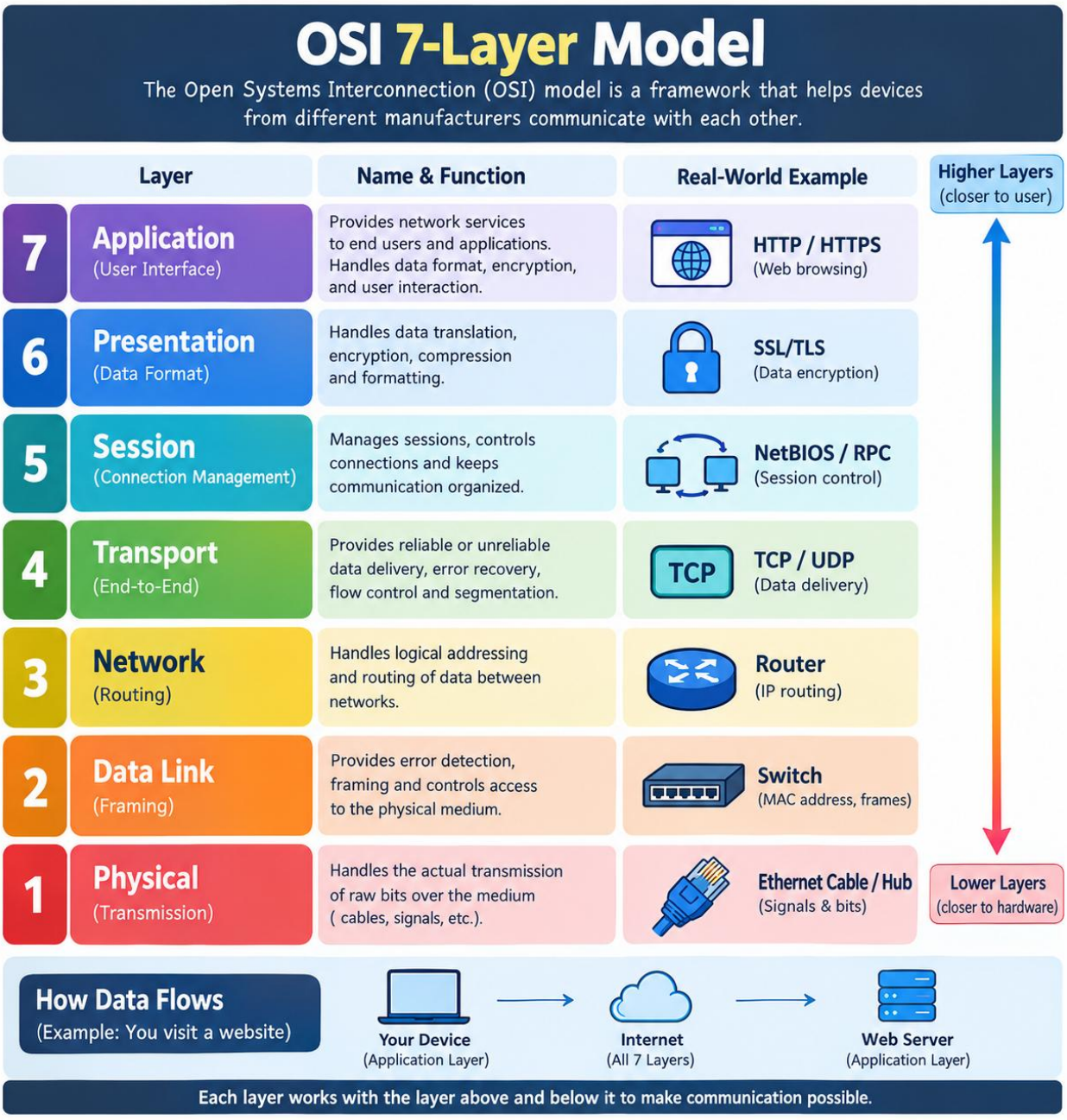
Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames. Write your answers as 'Scenario: Layer Name (Layer Number)'

a) Encrypting data before sending: Presentation Layer (Layer 6)

b) Routing packets across networks: Network Layer (Layer 3)

c) Converting data to electrical signals: Physical Layer (Layer 1)

d) Opening a web page in your browser: Application Layer (Layer 7)

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Example: WhatsApp

When we send a message on WhatsApp, different OSI layers work together:

1. Application Layer (Layer 7): We type a message and press the Send button in WhatsApp.

2. Transport Layer (Layer 4): It helps transfer the data between our phone and the WhatsApp server.

3. Network Layer (Layer 3): It uses IP addresses to route the data to the correct destination.

4. Data Link Layer (Layer 2): It transfers data through the local Wi-Fi or mobile network and checks for errors in frames.

5. Physical Layer (Layer 1): It sends data as electrical, radio, or other physical signals through the network.

In simple words: These layers work together to send our WhatsApp message from our phone to the receiver's phone.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

The problem may be in the Application Layer (Layer 7) or the Presentation Layer (Layer 6).

- **Application Layer (Layer 7): The web browser or HTTP/HTTPS service may not be working properly.**
- **Presentation Layer (Layer 6): There may be an encryption or SSL/TLS certificate problem.**

Reason: Since we can successfully ping the website's IP address, basic network connectivity is working. The problem may be with the website service, browser, or HTTPS connection.

Example: The server responds to ping, but the website does not open in Chrome.